DAI Yuhang, Peter

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Location: Hong Kong SAR, China

Education

- **The Hong Kong Polytechnic University** *Sep 2022 - Jun 2026 (Expected)*
  BSc (Hons) Computer Science with Minor in Applied Mathematics Hong Kong
  - **Overall GPA:** 3.74/4.00 (3.81/4.30)
  - **Major GPA (Computer Science):** 3.98/4.30
  - **Minor GPA (Mathematics):** 3.94/4.30
  - **Key Courses:** Computer Vision (A+), Big Data Analytics (A+), Data Structure (A+), System Programming (A+), Computer Security (A+), Software Engineering (A+), Operating System (A), Computer Networking (A), Database Systems (A), Artificial Intelligence of Things (A), Design and Analysis of Algorithms (A)
- **Tongji University** *Jul 2023*
  Summer School - Artificial Intelligence Greater Bay Area, China
  - Attended intensive lectures on cutting-edge AI technologies and industry applications
  - Collaborated on research report addressing smart healthcare solutions using AI

Research Experience

- **PolyU Yang Group** [] *Aug 2025 - Present*
  Research Assistant - PI/Supervisor: Prof. YANG Xingyi Hong Kong
  Project: VGGT-Dense2Sparse - Sparse-View 3D Reconstruction with Vision Transformer Fine-Tuning (Initial Stage)

  Tech Stack: PyTorch, PyTorch3D, LoRA, Vision Transformers, HuggingFace PEFT, Sinkhorn OT

  - Implemented asymmetric LoRA fine-tuning for VGGT Vision Transformer with adaptive rank configuration
  - Developed novel Sinkhorn optimal transport-based attention alignment loss replacing KL divergence, with teacher-student distillation framework enabling dense-to-sparse view knowledge transfer and achieving **+6% point cloud metrics** and eliminating performance plateau after epoch 18 (Initial Achievements)
- **PolyU PolySmart Group** [] *Dec 2024 - Sep 2025*
  Research Assistant - PI: Prof. LI Qing, Supervisor: Prof. WEI Xiaoyong Hong Kong
  Project: Interactive-IIVT - Interactive Video Retrieval with Neural Reranking []

  Tech Stack: PyTorch, CUDA, CLIP/BLIP Embeddings, Active Learning

  - Implemented neural network-based reranking model (2048→512→256 dims) with adaptive threshold classification, processing video embeddings from CLIP/BLIP for TRECVID AVS benchmarks (TV19/20/21 with 70 queries total)
  - Designed adaptive theta parameter mechanism to dynamically balance embedding-based and concept-based scoring through score separation analysis, with learning rate of 0.1 for iterative refinement across 20 interaction rounds
  - Developed active learning pipeline with uncertainty sampling (videos near 0.7 decision boundary) and bottom-rank negative sampling strategy, achieving balanced exploration-exploitation for user feedback collection
  - Achieved **17% improvement** over random baseline in cumulative video retrieval (244→285 videos), **Precision@400 of 0.71** (+54% gain), and outperformed BLIP2 by **97%** and CLIP2 by **164%**

Project: MolGround - Molecule Grounding Benchmark for Structure-Text Alignment [ ]

Tech Stack: RDKit, SMILES Parsing, PubChem API, Substructure Matching

- Constructed comprehensive benchmark dataset of **1,006 molecule grounding instances** with atom-level annotations linking natural language descriptions to precise molecular substructures via SMILES representations
- Developed multi-stage annotation pipeline including substructure extraction, chemical name normalization, SMILES mapping, and atom index grounding with visual verification for quality control
- Implemented sentence-level granularity supporting multi-instance grounding (e.g., mapping "four thiophene rings" to distinct atom groups) and hierarchical structure recognition from simple atoms to complex functional groups
- **Manually reviewed and corrected** all annotations with expert verification of grounding accuracy, name mappings, and completeness flags, ensuring high-quality training data for molecule-language models
- Enabled applications in molecule understanding, chemistry-language alignment, molecular retrieval, drug discovery, and fine-grained molecule-text evaluation benchmarks

- **PolyU AIoT Lab** [ ] — *Sep 2023 - May 2024*

  Research Assistant - PI: Prof. Prof. CAO Jiannong, Supervisor: Dr. HUANGFU Yaguang — Hong Kong

  Project: eLLM - Elastic LLM Inference Framework [ ] [ ]

  Tech Stack: Rust, AVX-512/AVX2 SIMD, FP16 Intrinsics, Inline Assembly, Intel Xeon (Sapphire Rapids)

  - Implemented 11 high-performance SIMD operators (MatMul, Softmax, GELU, SiLU, RMS Norm, etc.) using AVX-512 FP16 intrinsics, processing 32 half-precision elements per vector operation for LLaMA3-70B inference
  - Achieved 27% performance improvement in MatMul through 512-byte memory alignment (27.1ms → 19.8ms), with cache-optimized blocking and fused multiply-add operations outperforming GPU-based systems for million-token sequences
  - Designed all figures and tables for the research paper using draw.io, including system architecture diagrams, performance comparison charts, and operator workflow illustrations
  - Developed automated testing infrastructure with benchmarks for continuous performance monitoring across different SIMD variants (AVX-512, AVX2)

Selected Research Projects

- **LogoCleaner - AI-Powered Logo Detection and Removal System** — *Jan 2025*

  Python, SAM, U-Net, CycleGAN, Flask, Vue.js, DALL-E 2 [ ]

  - Implemented three approaches: SuperPixel+RF baseline, U-Net deep learning, and state-of-the-art SAM+NN
  - Achieved 0.29 average dice loss with 5-second detection and 15-second inpainting latency
  - Developed full-stack web application for real-time logo removal with Vue.js and Flask

- **Synth2Det - Domain Adaptation for Adverse Weather Object Detection** — *Jan 2025*

  Python, CycleGAN, YOLOv11, PyTorch, Weights & Biases [ ]

  - Implemented CycleGAN for realistic clear-to-snow weather translation on driving scenes
  - Developed 3-phase training strategy with test-time adaptation for robust detection
  - Demonstrated significant performance gains over baseline on synthetic validation datasets

- **PolySecure - End-to-End Encrypted File Storage System** — *Apr 2025*

  Python, AES-256-GCM, RSA-2048, TOTP, SQLite, TLS/SSL [ ]

  - Architected secure file storage with client-side encryption and granular access control
  - Implemented comprehensive security features including 2FA, secure key exchange, and digital signatures

Industry Experience

- **Sina Corporation** [ ] — *May 2024 - Jul 2024*

  Technical Department Intern - Supervisor: Mr. LIANG Lilong — Beijing, China

  - Developed automated web crawling systems using Selenium and Scrapy for large-scale data collection
  - Utilized GPU infrastructure to deploy LLMs for data generation and classification tasks

Honors and Awards

- **Dean's List, Faculty of Engineering** *2023/24, 2024/25*
  The Hong Kong Polytechnic University
  - Awarded for maintaining GPA above 3.7 for consecutive academic years
- **Department of Computing Scholarship for Hall Residents** *May 2025*
  PolyU Student Affairs Office and COMP Department
  - HKD 6,000 scholarship recognizing academic excellence and extracurricular involvement

Leadership and Service

- **Committee Member** *Jul 2024 - May 2025*
  PolyU Dreaming Cameramen Club
  - Organized 5 major photography events with over 120 participants each
- **Hall Activity Organizer** *Sep 2023 - May 2025*
  PolyU Student Halls
  - Coordinated 10 cultural and educational events serving 80+ residents each
  - Served as Master of Ceremonies for multiple hall events
- **Community Service Volunteer** *Sep 2024 - May 2025*
  Various Organizations in Hong Kong [Certificate]
  - Contributed 80+ hours in cultural exchange, autism support, and environmental initiatives

Professional Memberships

- **Photographer Specialized Skills Certification**, Membership ID: CGCC311725002793 *2024 - Present*
  - completed the unilied training and passed the tests required by China General Chamber of Commerce.

Technical Skills

- **Programming Languages**: Python, C/C++, Rust, Java, JavaScript, LaTeX
- **Deep Learning Frameworks**: PyTorch, Hugging Face Transformers
- **Computer Vision Libraries**: OpenCV, CLIP, SAM, Detectron2, MMDetection
- **Tools and Platforms**: Git, Docker, CUDA, Weights & Biases, Linux, AWS